

# 18 Data

**4 primary questions** | Oldest to newest

2026 objectives: 3.6

20 related questions are filed elsewhere in this volume. Find them by selecting this chapter and enabling related topics in [index.html](#).

Click a question below to begin.

|                                              |          |
|----------------------------------------------|----------|
| <a href="#">CA1-P2   April 2012   Q5</a>     | 16 marks |
| <a href="#">CA1-P2   April 2013   Q6</a>     | 20 marks |
| <a href="#">CA1-P2   September 2018   Q3</a> | 13 marks |
| <a href="#">CP1-P2   September 2024   Q2</a> | 48 marks |

**CA1-P2 | April 2012 | Q5**

Current topic | 16 marks total | Original PDF p3

Related: Ch 20.

- 5** A medium sized manufacturing and distribution company currently has no formal provisions for sick pay. The written contractual employment terms are that if an employee is unable to work due to sickness the employee will not be paid by the employer.

The company operates an informal, discretionary policy where, depending on circumstances and on the decision of their immediate manager, some employees do receive some sick pay. The level and duration of such payments varies and can be changed or stopped with minimal notice.

- (i) Outline the possible rationale behind the company's existing arrangements for sick pay. [4]

The company is now proposing to introduce formal sick pay arrangements.

Employees who are absent due to sickness will receive full pay from the company for five working days without the need to produce evidence of sickness.

After that time, if certification of sickness from a doctor is provided, payments will be made as a percentage of full pay and the percentage will be as follows:

First three months 100%

Next three months 50%

Thereafter, payments will cease and no discretionary payments will be made under any circumstances.

- (ii) Explain why the company's recent sickness experience may not be suitable for estimating future sickness experience and hence payments arising under the new arrangements. [9]
- (iii) Set out difficulties the company may face were it to try to use any other sources of experience data for sickness rates. [3]

[Total 16]

**CA1-P2 | April 2013 | Q6**

Current topic | 20 marks total | Original PDF p3

Related: Ch 37.

- 6** (i) List the main uses of data in actuarial work. [4]

A defined benefit pension scheme is closed to future benefit accrual. Its latest annual valuation has just been completed and this shows that the scheme's surplus has increased to ten times its previous level.

- (ii) Outline the checks on the data that should be made to ensure the surplus is correct. [7]

The data has been checked and no errors have been found.

- (iii) Suggest reasons why the surplus could have increased. [5]

One of the trustees suggests that the scheme should take advantage of the increase in surplus, by winding up the scheme and using the surplus to improve members' benefits.

- (iv) Comment on this suggestion. [4]  
[Total 20]

**CA1-P2 | September 2018 | Q3**[Current topic](#) | 13 marks total | [Original PDF p2](#)

September 2018 main paper, recovered from the supplied archive; the extracted folder had contained the spare paper under this date. Related: Ch 22.

- 3** ABC is a large company. It is considering the purchase of XYZ, a much smaller company which operates in the same industry. Both companies offer their employees a defined benefit pension scheme.

ABC is considering whether, as part of the purchase, it would be practical to take over the XYZ pension scheme and merge it with its own scheme.

- (i) Outline the information that ABC would need to calculate the cost of taking over the XYZ pension scheme. [4]
- (ii) Describe the checks that will need to be made on the information in part (i). [4]
- (iii) Discuss the issues ABC will need to take into account before proposing the merger of the two schemes. [5]

[Total 13]

**CP1-P2 | September 2024 | Q2**

Current topic | 48 marks total | Original PDF p3

Integrated case study. The full scenario is retained under one primary chapter; related chapters identify its other substantial topics. Related: Ch 17, Ch 19, Ch 29, Ch 22, Ch 33, Ch 28.

- 2** A large company that sponsors a large benefit scheme for its employees, has recently purchased a number of companies, which each had small employee benefit schemes. These schemes are currently managed separately.

The various scheme managers are undertaking several significant projects.

- Project A: Updating an existing benefits calculation model to use an automatic market data feed rather than rely on manual inputs of the data.
  - Project B: Improving the quality of membership data held on its systems by writing to members to obtain more information about them and their dependants.
  - Project C: Transferring a proportion of one of the scheme's liabilities to an insurance company. In return for a single premium, the insurance company will take on the liability for making future benefit payments to the transferring members. The premium will be paid based on initial membership data, with both parties having 12 months to agree final data. The final premium payable will be based on finalised data and the insurance company will determine whether the scheme needs to pay additional funds or not.
  - Project D: Merging all of the employee benefit schemes into one scheme.
  - Project E: Creating a member website where all information about the schemes and the benefits for each member can immediately be accessed.
- (i) Outline the criteria that the scheme managers should use to prioritise the five projects set out above. [3]
  - (ii) Discuss the rationale for Project A. [5]
  - (iii) Outline how the scheme can test that the benefits calculation model still works after Project A has been implemented. [5]
  - (iv) Outline possible issues with gathering data about the members and their dependants in Project B. [5]
  - (v) Describe how the scheme could allow for the results from Project B when setting its demographic assumptions. [5]
  - (vi) Discuss the advantages and disadvantages of the transfer in Project C. [5]
  - (vii) Discuss the possible risks to the insurance company of agreeing the transaction in Project C based on initial membership data. [4]
  - (viii) Discuss the advantages and disadvantages to the company of merging the schemes in Project D. [5]

## CP1-P2 | September 2024 | Q2 - continued

[Current topic](#) | 48 marks total | [Original PDF](#) p4

Integrated case study. The full scenario is retained under one primary chapter; related chapters identify its other substantial topics. Related: Ch 17, Ch 19, Ch 29, Ch 22, Ch 33, Ch 28.

- (ix) Outline the information that should be provided to members in Project E in respect of:
    - (a) the schemes.
    - (b) their own benefits.

[7]
  - (x) Outline how a risk reporting structure would assist the scheme managers. [4]
- [Total 48]